

The kitchen of the future: Testing user experience through virtual reality

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The evolution of virtual reality (VR) has led to a range of innovative applications in various fields, and the culinary household appliance industry is no exception. In our pursuit of exploring new possibilities for the kitchen of the future and within the context of a collaborative research project with a leading company in the sector like BSH, we have posed the question: How can we harness virtual reality (VR) to enhance user testing processes?

This contribution aims to explore the potential of VR as an innovative tool to enhance evaluation processes in the development of the kitchen of the future. We believe that our research can contribute to understanding how VR can influence user interactions with products and ultimately drive innovation and continuous improvement in the field of user experience (UX) testing for the kitchen of the future.

Introduction

User experience (UX) testing is paramount in the development of new products and services. Traditionally, market research studies and user testing have relied on conventional methods such as questionnaires, face-to-face interviews and direct observation. However, these approaches may have limitations in terms of reaching a wide range of users, conducting test across different countries, precisely understanding user attention during interactions with products or the substantial expense and effort involved in constructing a physical prototype, which may be completely rebuilt in the event of any necessary modifications.

VR provides a unique opportunity to overcome these limitations by offering an immersive, portable and, to a greater extent, automatable environment. Although there are currently

challenges regarding interaction in virtual environments, we believe that VR can bring valuable improvements to evaluation processes.

In this contribution, we present our approach to utilizing VR in UX testing within the context of the kitchen of the future. We explore the advantages and limitations of this technology in evaluating UX and describe the experimental design and preliminary results of our study. Additionally, we discuss the practical implications and potential future research directions in this emerging field.

Methodology

A study was conducted based on a test previously conducted by the company BSH, which involved a medium level of interaction between the user and the product. The test involved evaluating five types of controls for a new induction cooktop solution. In this test, there was physical/tactile interaction with the prototypes, although they were not fully functional since the goal was not to test the functionality of each prototype but to understand user preference. This aspect is crucial as we needed to explore how to address or transfer this interaction to VR in this initial approach, without the physical prototypes of the controls.

The primary objective of the study is to ascertain the feasibility of conducting the test in VR. For this purpose, an initial pilot test was carried out and once the viability of the test was confirmed, we design the final test.

In this initial pilot of the test, the exact same test conducted by BSH was replicated in VR, modeling the same space as the original test and reproducing the prototypes with their original manufacturing flaws.

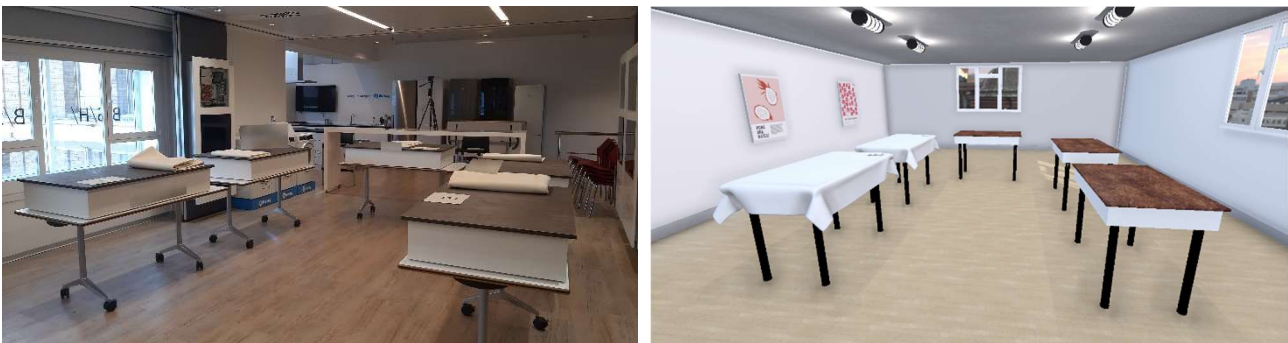


Figure 1. Left: representation of the test conducted by BSH. Right: virtual reality modelling of the same test.

During the test design, the importance of combining real physical supports during the use of VR headsets was recognized, allowing users to rest on a tangible surface that matched the one seen through the goggles, promoting a secure interaction. Additionally, it was necessary to acquaint participants, especially those unfamiliar with VR, with wearing and using VR devices. In order to ensure consistent results, the users were first invited to freely explore a different scenario from the test and once they were familiarized, then they saw the test environment.

In the final test, the advantages offered by virtual reality were leveraged, such as creating a realistic kitchen immersive environment, improvements and updates to the prototypes (e.g., interfaces), which could not be carried out in the original test due to manufacturing costs and time constraints.



Figure 2. Immersive environment of the final test.

The original test methodology was divided into four blocks, which were carried out in the same way in both VR and the original test, involving constant dialogue between the user and the interviewer.

The first block focused on the user's lifestyle in relation to their cooking habits. The second block addressed how the user perceives the verbally described product and its ideal control. The third block involved the interaction phase, and the last block consisted of an evaluation. It was in this third block that the user put on the VR glasses and carried out the UX test until its completion. Finally, a fifth block was added in which, instead of presenting the controls one by one, all of them were shown simultaneously to encourage creative brainstorming and ensure that no potential solution was overlooked.



Figure 2. Left: diagram of the dynamic of the test. Right: real picture of the test.

Results and conclusion

The qualitative results obtained from the tests indicate that the identified pains and gains for each concept follow a similar trend, revealing that the concepts are understood in the same way in both reality and VR. This confirms the viability of conducting UX tests in VR. The first quantitative results, reported with a sample size of ten users, indicate that preferences remain consistent, with the same favorite and least favorite concepts, and the remaining concepts maintain the same order of preference with a single exception. We hypothesize that this could be attributed to various factors, such as differences in user characteristics, among others, which could be explored in future research.

Regarding the overall findings, we observe the brain's ease of adaptation. Users had a genuine sense of immersion in the environment shown to them. Furthermore, having a physical support helps enhance the immersive experience and user safety. The physical support worked even if its height was not an exact match to what was seen through the goggles. Initially, users noticed the difference, but they adapted within seconds once they touched it. Along the same line, the training to become familiar with the devices took less than a minute in this case. On the other hand, people did not experience dizziness or fatigue, a critical consideration given the potential challenges in VR experiments. Lastly, in some cases, hearing the interviewer without seeing them led to confusion. This aspect is open for further exploration in future studies.

In conclusion, both the Lab and BSH agreed on the viability of conducting VR tests and its potential advantages, such as the ease and cost-effectiveness of modifying virtual prototypes. Moreover, it provides the opportunity to conduct tests in various locations and reach a broader audience more easily. Since virtual reality is still in the process of improvement and development, and has not yet reached the point where it can be fully automated to the extent that anyone with VR goggles can perform a test without the assistance of an interviewer, our future work will focus on exploring ways to automate VR testing, determining the optimal levels of interaction for its use, and implementing interaction enhancements, such as hand tracking.

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